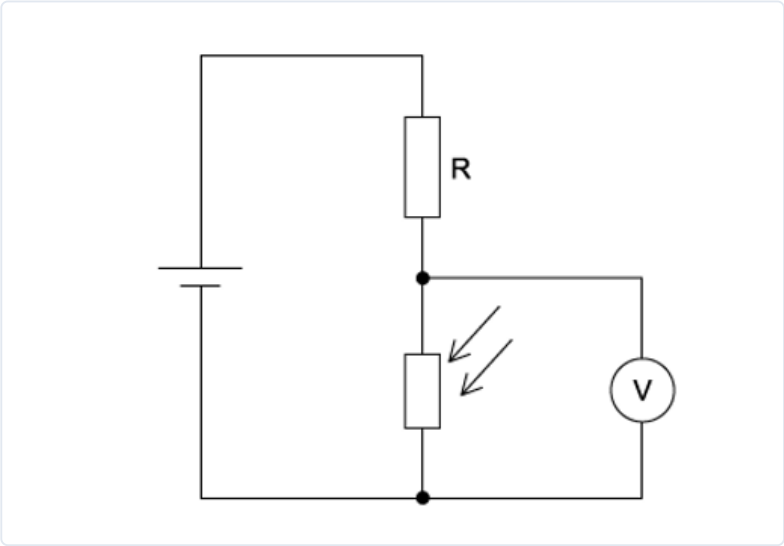
CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

10.3 Potential Dividers

Syllabus Section: 10. D.C. Circuits **Paper Type:** Multiple Choice (Paper 1) **Total:** 29 Questions (29 Marks)

Question 1 Easy [1 mark]

A potential divider consists of a fixed resistor R and a light-dependent resistor (LDR)



What happens to the voltmeter reading, and why does it happen, when the intensity of light on the LDR increases?

A The voltmeter reading decreases because the LDR resistance decreases

B The voltmeter reading decreases because the LDR resistance increases

C The voltmeter reading increases because the LDR resistance decreases

D the voltmeter reading increases because the LDR resistance increases

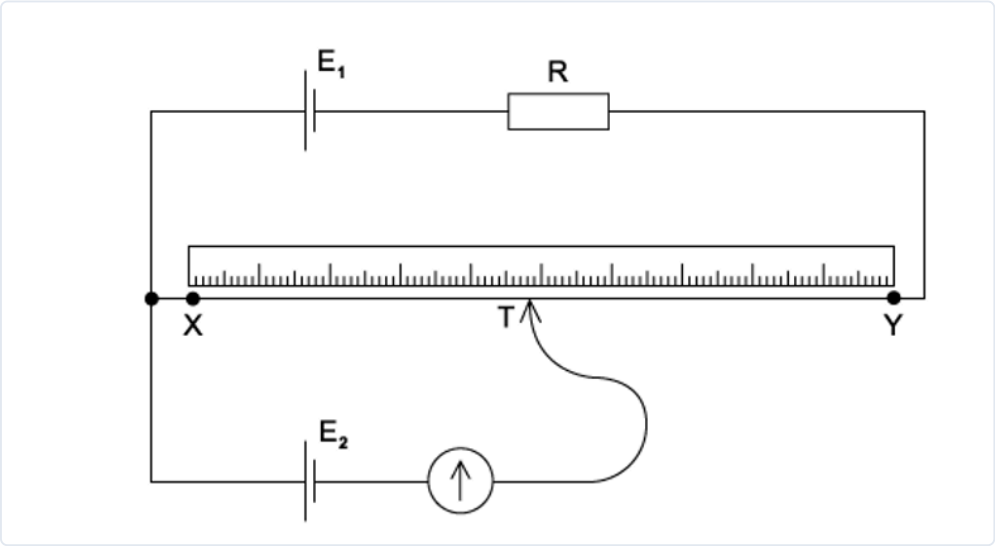
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The resistance of the LDR decreases as the light intensity increases
- Ohm's law states $V = IR$
- Since the resistance of the LDR decreases, the potential difference across it also decreases
- Therefore, as the LDR resistance decreases, the voltmeter reading decreases

Question 2 Easy [1 mark]

The diagram shows a potentiometer circuit.



The contact T is placed on the wire and moved along the wire until the galvanometer reading is zero. The length XT is then noted.

In order to calculate the potential difference per unit length of the wire XY, which value must also be known?

A the e .m .f. of the cell E_1

B the e. m .f. of the cell E_2

C the resistance of resistor R

D the resistance of the wire XY

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- When the galvanometer reading is zero, this means that the p.d. across XT of the wire is the same as the e.m.f. E_2 connected to the galvanometer
- To calculate the potential difference per unit length on the wire XY, we need to know the length of the wire, and the p.d. across it
- The p.d. across XT is equal to the e.m.f. E_2 when the galvanometer reading is zero

- So, we need to know the e. m .f. of the cell E_2 to know the p. d. across the wire

Question 3 Easy [1 mark]

A source of e.m.f. 9.0 mV has an internal resistance of 6.0 Ω.

It is connected across a galvanometer of resistance 30 Ω.

What is the current in the galvanometer?

A 250 μA

B 300 μA

C 1.5 mA

D 2.5 mA

Mark Scheme & Explanation:

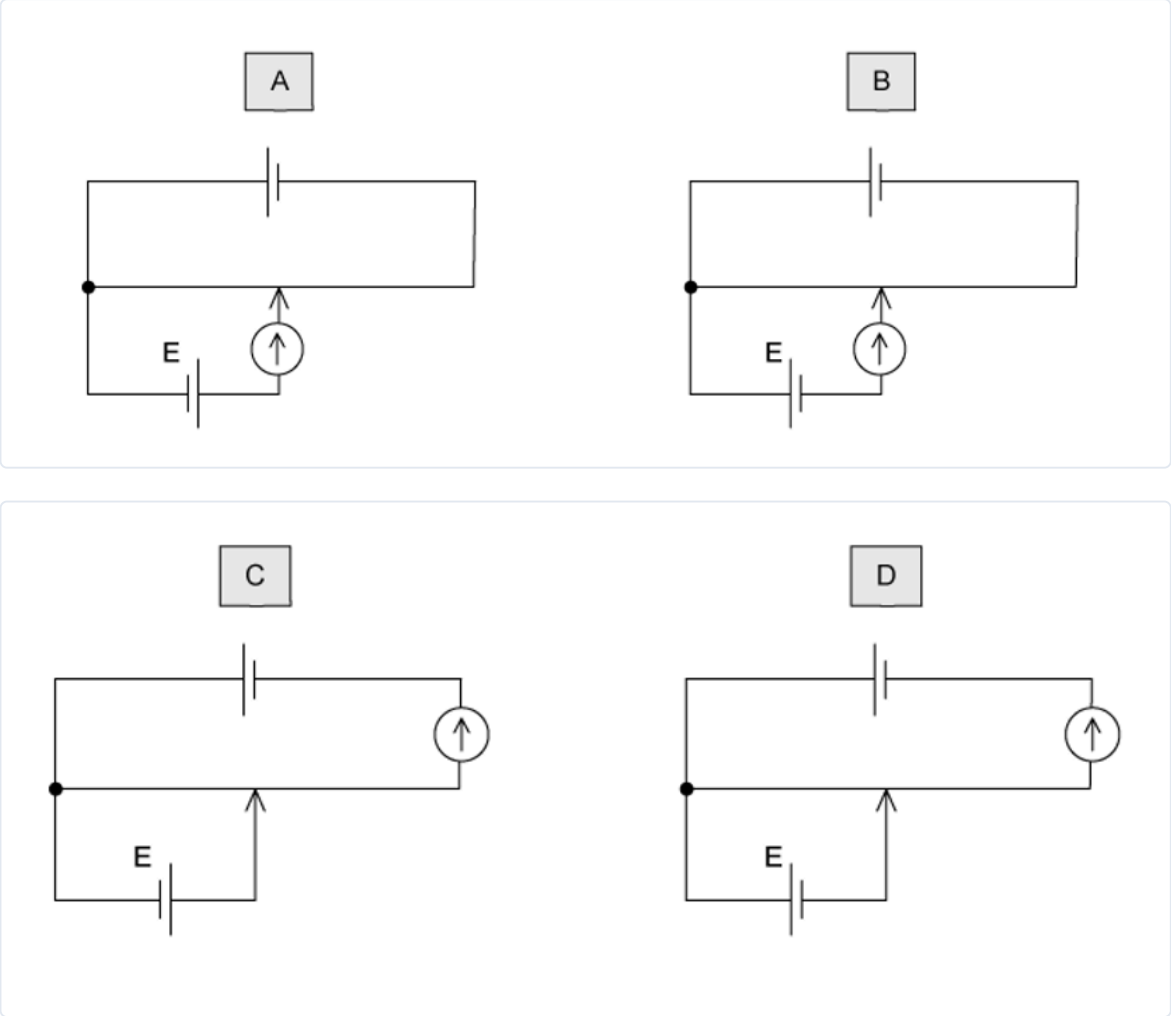
*The correct answer is **A** because:*

- The galvanometer and the internal resistance of the cell can be treated as two resistors in series
- So, the total resistance *R* in the circuit = 6 + 30 = 36 Ω
- Ohm's law states $V = IR$
  - So, current $I = \frac{V}{R} = \frac{9.0 \times 10^{-3}}{36} = 2.5 \times 10^{-4} \quad A = 250 \times 10^{-6} \quad \text{A}$
- In a series circuit, the same current flows through the cell and the galvanometer
  - Therefore, the current in the galvanometer is 250 μA

Question 4 Easy [1 mark]

The unknown e.m.f. *E* of a cell is to be determined using a potentiometer circuit. The balance length is to be measured when the galvanometer records a null reading.

What is the correct circuit to use ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- For the galvanometer to give a null reading, the potential difference across it should be zero
- This can only occur when the position at which it is connected on the wire gives a potential difference (on the wire) equal to the cell of e .m .f. *E*

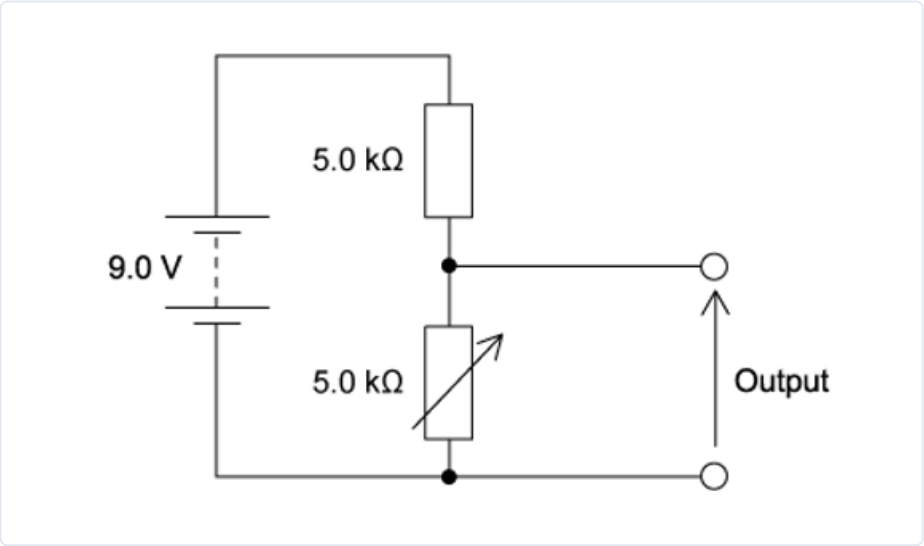
- The cell should be connected such that its potential opposes the potential on the wire
- When the potential difference across the galvanometer is zero, no current flows through it, and the galvanometer records a null reading
    - This set up is demonstrated by circuit **B**

A is incorrect because current flows from the positive terminal to the negative terminal, so, the positive terminal of the cell cannot be connected directly to the galvanometer because this will always cause a current to flow through it

C & **D** are incorrect because the galvanometer cannot be connected directly to the negative terminal of the main supply because all the current would return to the negative terminal of the main supply, causing the galvanometer always to give a non-zero reading

Question 5 Easy [1 mark]

The diagram shows a potential divider circuit designed to provide a variable output p. d.



Which gives the available range of output p. d ?

	Maximum output	Minimum output
A	3.0 V	0
B	4.5 V	0
C	9.0 V	0
D	9.0 V	4.5 V

A

B

C

D

Mark Scheme & Explanation:

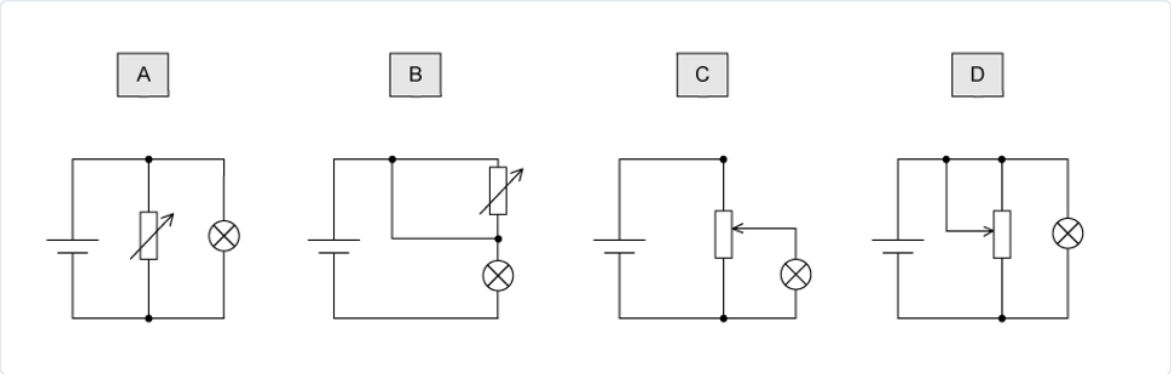
*The correct answer is **B** because:*

- The potential divider equation is given by
  - $V_{out} = V_{in} \times \frac{R_1}{R_1 + R_2}$
  - Where R_1 = 5 kΩ, R_2 = 5 kΩ and V_{in} = 9.0 V
  - $V_{out} = 9 \times \frac{5}{5+5} = 4.5 \text{ V}$

- Therefore, the minimum output of the variable resistor is 0 V, and the maximum output is 4.5 V

Question 6 Easy [1 mark]

Which diagram shows a potential divider circuit that can vary the voltage across the lamp ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

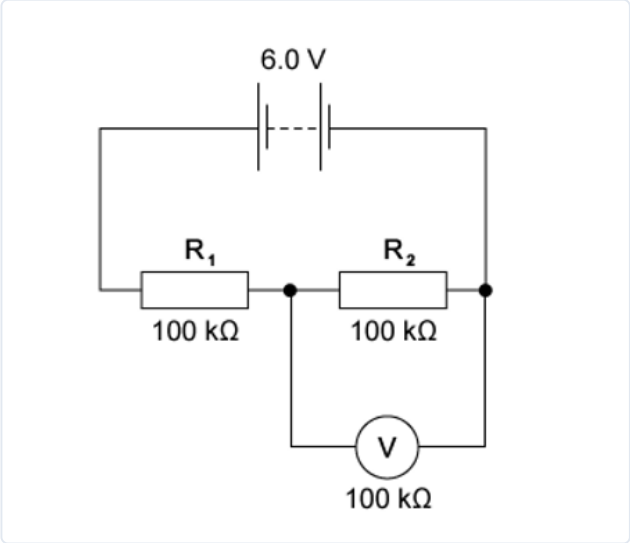
- A variable potential divider (potentiometer) can be used to vary the brightness of a light bulb between zero and normal brightness
- The moveable contact, represented by the arrowhead, splits the resistor in two allowing it to act as though there were two resistors in the circuit and hence like a variable potential divider or potentiometer
- Changing the resistance of the potentiometer changes the potential difference of the lamp
- This is represented by circuit **C** as the lamp needs to be directly connected to the potentiometer

A & **B** are incorrect because a potentiometer is represented by an arrow pointing to a resistor

D is incorrect because the wire with the arrowhead needs to connect the lamp directly to the potentiometer in order to vary the potential difference of the lamp

Question 7 Easy [1 mark]

In the circuit shown, the 6.0 V battery has negligible internal resistance. Resistors R_1 and R_2 and the voltmeter have resistance 100 kΩ.



What is the current in the resistor R_2?

A 20 μA

B 30 μA

C 40 μA

D 60 μA

Mark Scheme & Explanation:

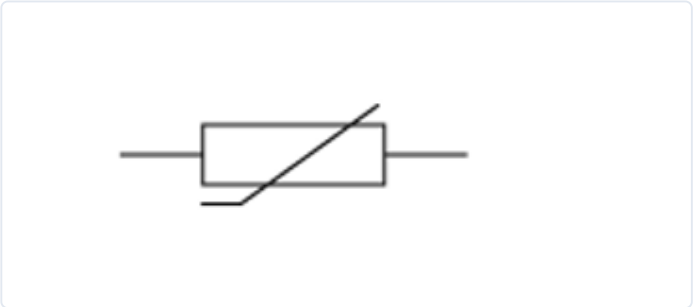
*The correct answer is **A** because:*

- We can treat the three components as two sets of components in series (R_1 and the parallel combination of R_2 and the voltmeter)
- So, the total resistance in the circuit $= 100 + \frac{1}{100} + \frac{1}{100}^{-1} = 100 + 50 = 150 \quad \text{k}\Omega$
- Ohm's law states $V = IR$
  - Therefore, the total current in circuit $I = \frac{6.0}{150} \quad = \quad 40 \quad \mu\text{A}$
- This current of 40 μA flows through resistor R_1, but it splits at the junction of R_2 and the voltmeter, which are in parallel

- Since both have the same resistance, the current splits equally
    - So, the current in $R_2 = \frac{40}{2} = 20 \quad \mu A$

Question 8 Easy [1 mark]

Which electrical component is represented by the following symbol ?



A a diode

B a potentiometer

C a galvanometer

D a thermistor

Mark Scheme & Explanation:

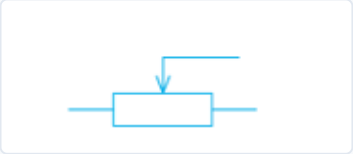
*The correct answer is **D** because:*

- This symbol represents a thermistor

A is incorrect because the symbol for a diode is



B is incorrect because the symbol for a potentiometer is

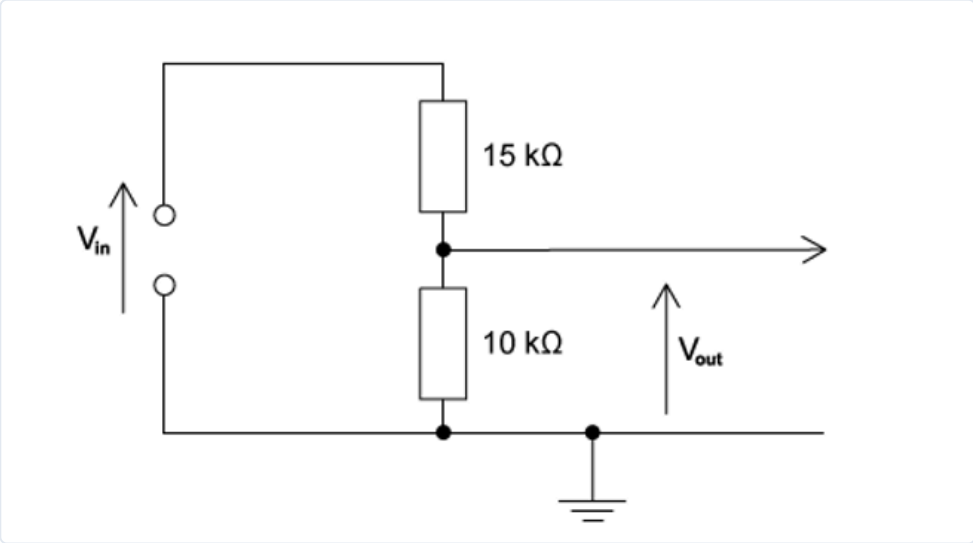


C is incorrect because the symbol for a galvanometer is



Question 9 Easy [1 mark]

The circuit is designed to trigger an alarm system when the input voltage exceeds some preset value. It does this by comparing V_{out} with a fixed reference voltage, which is set at 4.8 V .



V_{out} is equal to 4.8 V.

What is the input voltage V_{in}?

A 4.8 V

B 7.2 V

C 9.6 V

D 12 V

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The potential divider equation is given by
- $V_{out} = V_{in} \times \frac{R_1}{R_1 + R_2}$
  - Where R_1 = 10 kΩ, R_2 = 15 kΩ and V_{out} = 4.8 V
- $V_{in} = V_{out} \div \frac{R_1}{R_1 + R_2} = V_{out} \times \frac{R_1 + R_2}{R_1}$
- $V_{in} = V_{out} \frac{R_1 + R_2}{R_1} = \frac{4.810 + 15}{10} = 12 \quad \text{V}$

- Therefore, the input voltage is 12 V

Question 10 Easy [1 mark]

A low-voltage supply with an e. m. f. of 20 V and an internal resistance of 1.5 Ω is used to supply power to a heater of resistance 6.5 Ω in a fish tank.

What is the power supplied to the water in the fish tank ?

A 41 W

B 50 W

C 53 W

D 62 W

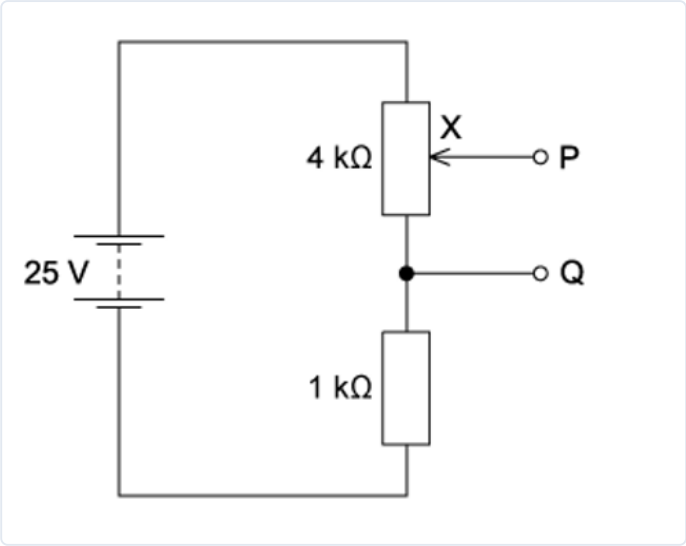
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The power supplied to the heater is $P = VI = \frac{V^2}{R}$
- From the potential divider equation:
  - $V_{heater} = V_{e.m.f} \times \frac{R_1}{R_1 + R_2}$
  - $V_{heater} = 20 \times \frac{6.5}{6.5+1.5} = 16.25 \quad \text{V}$
- So, $P = \frac{V^2}{R} = \frac{16.25^2}{6.5} = 40.625 \quad \text{W}$
- Therefore, the power supplied to the heater is 41 W

Question 11 Medium [1 mark]

The diagram shows a potential divider circuit which, by adjustment of the contact X, can be used to provide a variable potential difference between the terminals P and Q.



What are the limits of this potential difference?

A 0 and 5 V

B 0 and 20 V

C 0 and 25 V

D 5 V and 25 V

Mark Scheme & Explanation:

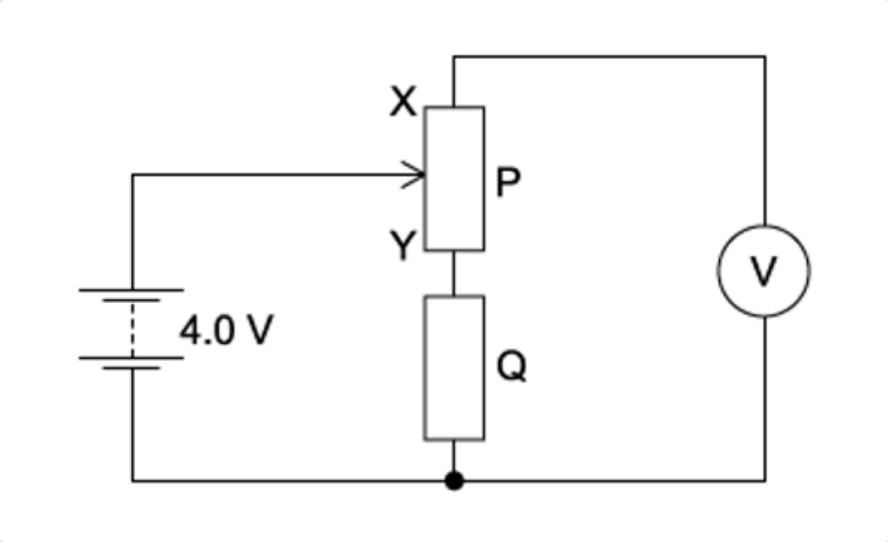
*The correct answer is **B** because:*

- The potential divider equation is given by
  - $V_2 = V_{e.m.f} \times \frac{R_1}{R_1 + R_2}$
- Let the resistance of the variable resistor be R_1 and let the resistance of the fixed resistor be R_2
  - R_1 varies from 0- 4 kΩ, R_2 = 1 kΩ and the e. m .f is V = 25 V
- The potential difference between the terminals P and Q is V_2
- The p. d. across a component is proportional to its resistance $V = IR$
  - So, the higher the resistance, the greater the p. d.

- For V_2 to be maximum, the resistance of the variable resistor, R_1 should be 4 kΩ
    - So the maximum p. d. is $V_2 = 25 \times \frac{4}{4+1} = 20 \quad V$
- For V_2 to be minimum, the smallest resistance is chosen, so, R_1 is zero
    - So, the minimum p. d. is $V_2 = 25 \times \frac{0}{0+1} = 0 \quad V$

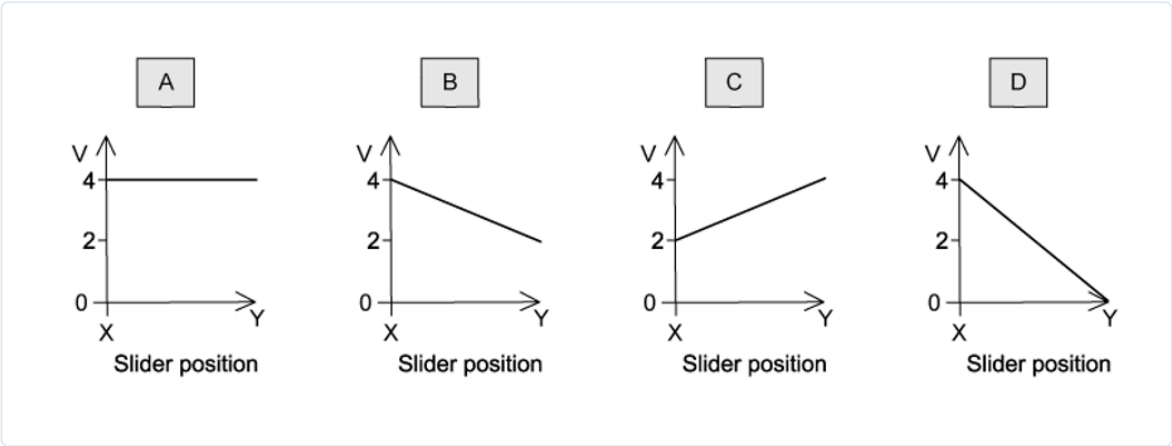
Question 12 Medium [1 mark]

In the circuit below, P is a potentiometer of total resistance 10 Ω and Q is a fixed resistor of resistance 10 Ω. The battery has an e .m f. of 4.0 V and negligible internal resistance. The voltmeter has a very high resistance .



The slider on the potentiometer is moved from X to Y, and a graph of voltmeter reading *V* is plotted against slider position.

Which graph would be obtained ?



A

B

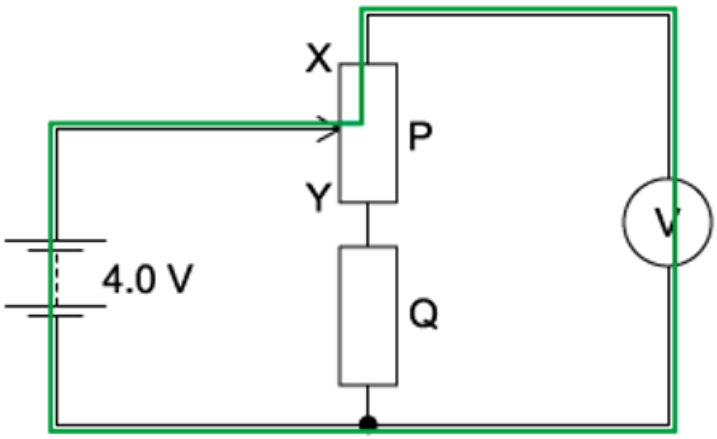
C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

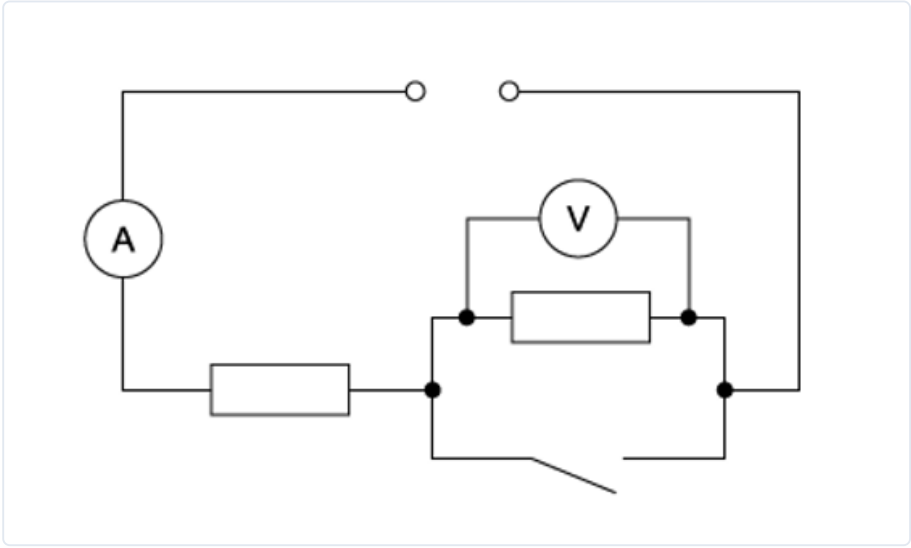
- When the slider is at X, where the resistance of the potentiometer is taken to be zero, the voltmeter must read 4 V
- The current flows: from the positive terminal — through the moving jack until the point it is connected on resistor P — from P to end X — through the voltmeter and back to the negative terminal of the supply, as shown in the diagram below
    - Note at no point does the current go through Q
- When the slider is moved to Y, the potentiometer P and the voltmeter are in series
- Still, no current flows through Q — resistor Q can be said to be short-circuited in both cases, with almost all the resistance in the voltmeter and hence the p.d. across the voltmeter is still 4 V
- Therefore, the voltmeter reading remains 4 V as the slider is moved from X to Y



Exam Tip: In a properly set up circuit, a voltmeter is usually connected in parallel to the component we want to know about. Connecting a voltmeter in series results in the above situation where the voltmeter always reads 4 V

Question 13 Medium [1 mark]

In the circuit below, the ammeter reading is *I*, and the voltmeter reading is *V*.



When the switch is closed, which row describes what happens to *I* and *V*?

	I	*V*
A	decreases	decreases to zero
B	increases	stays the same
C	increases	decreases to zero
D	stays the same	increases

A

B

C

D

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

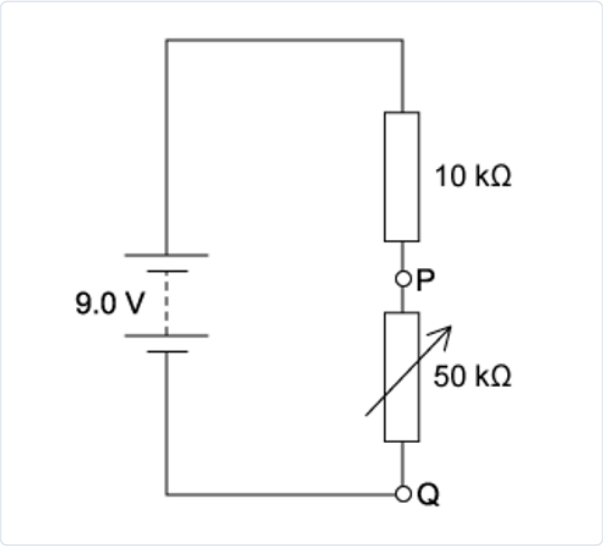
- When the switch is opened, current passes through both resistors in the circuit
- Let the resistance of one resistor be R, and we will assume they are of the same resistance
    - When the switch is opened, the total resistance R_T in the circuit = $R + R = 2R$
- Ohm's law states $V = IR$

    - This gives a current $I = \frac{E}{R_T}$ where E is the e .m. f. of the supply

- The total e .m .f in the circuit is divided as p. d. across each of the resistor
- When the switched is closed, the resistor connected in parallel to the wire is short-circuited
- That means, all current would pass through the wire (of negligible resistance) instead of passing through that resistor
    - When the switch is closed, the total resistance R_T in circuit = R + 0 = R
- This gives a current I which is now greater than before, since R_T is less
- Since the current I through the resistor in parallel is now zero, the p .d. across it now also decreases to zero

Question 14 Medium [1 mark]

The diagram shows a potential divider connected to a 9.0 V supply of negligible internal resistance.



What range of voltages can be obtained between P and Q?

A zero to 1.5 V

B zero to 7.5 V

C 1.5 V to 7.5 V

D 1.5 V to 9.0 V

Mark Scheme & Explanation:

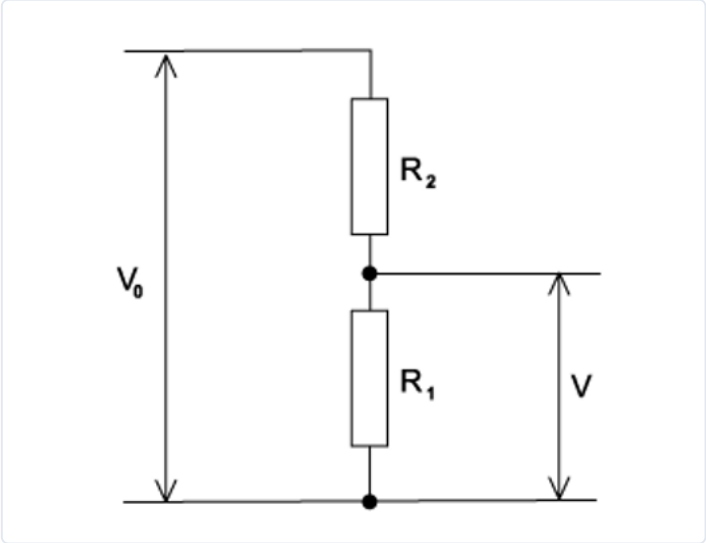
*The correct answer is **B** because:*

- The potential divider equation is given by
  - $V_2 = V_{e.m.f} \times \frac{R_1}{R_1 + R_2}$
- Let the resistance of the variable resistor be R_1 and let the resistance of the fixed resistor be R_2
  - R_1 varies from 0 - 50 kΩ, R_2 = 10 kΩ and the e. m. f is V = 9.0 V
- The potential difference between the terminals P and Q is V_2
- The p .d. across a component is proportional to its resistance $V = IR$
  - So, the higher the resistance, the greater the p. d.

- For V_2 to be maximum, the resistance of R_1 should be 50 kΩ
    - So the maximum p .d. is $V_2 = 9 \times \frac{50}{50+10} = 7.5 \quad \text{V}$
- For V_2 to be minimum, the smallest resistance is chosen, so, R_1 is zero
    - So, the minimum p. d. is $V_2 = 9 \times \frac{0}{10+0} = 0 \quad \text{V}$
- Therefore, the minimum output of the variable resistor is 0 V, and the maximum output is 7.5 V

Question 15 Medium [1 mark]

A potential divider consisting of resistors of resistance R_1 and R_2 is connected to an input potential difference of V_0 and gives an output p .d. of V.



What is the value of V ?

A $\frac{V_0 \quad R_1}{R_2}$

B $\frac{V_0 \quad R_1}{R_1 + R_2}$

C $\frac{V_0 \quad R_2}{R_1 + R_2}$

D $\frac{V_0 R_1 + R_2}{R_1}$

Mark Scheme & Explanation:

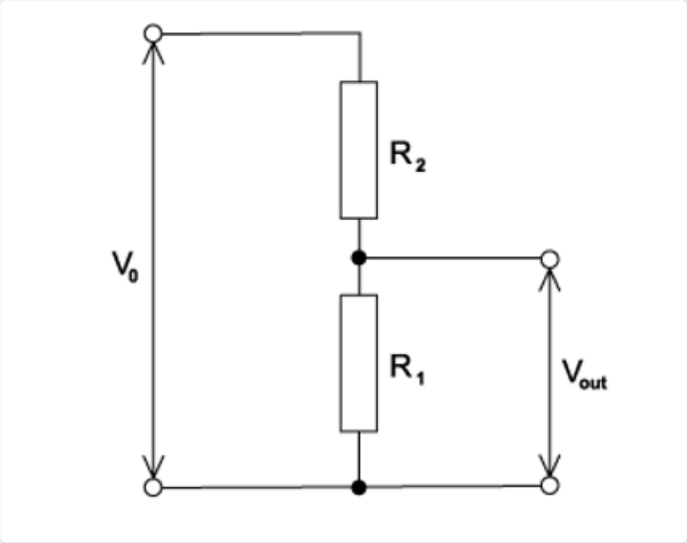
*The correct answer is **B** because:*

- We want to find the proportion of output p. d. which is equal to the p. d. across resistor R_1
- For this reason, the R_1 is on the top of the fraction in the potential divider equation
- The potential divider equation is given by
  - $V = V_0 \times \frac{R_1}{R_1 + R_2}$
- Therefore, V is equal to

- $V = \frac{V_0 \quad R_1}{R_1 \quad + R_2}$

Question 16 Medium [1 mark]

A potential divider consists of resistors of resistance R_1 and R_2 connected in series across a source of potential difference V_0. The potential difference across R_1 is V_{out}.



Which changes to R_1 and R_2 will increase the value of V_{out}?

	R_1	R_2
A	doubled	doubled
B	doubled	halved
C	halved	doubled
D	halved	halved

A

B

C

D

Mark Scheme & Explanation:

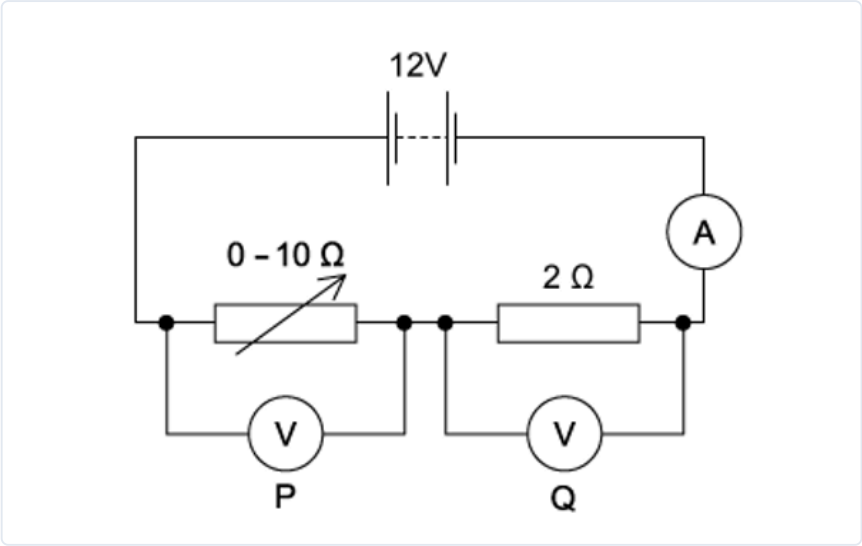
*The correct answer is **B** because:*

- The potential divider equation is given by
  - $V_{out} = V_0 \times \frac{R_1}{R_1 + R_2}$

- In order to increase the value of V_{out}, the value of R_1 should be maximized, while the value of R_2 should be minimized as:
  - $V_{out} \alpha \quad R_1$ and $V_{out} \quad \alpha \quad \frac{1}{R_1 + R_2}$
- Consider option **B**
  - $V_{out} \quad \alpha \quad \frac{2R_1}{2R_1 + \frac{1}{2}R_2}$
- Let's use some test values. If R_1 = 10 kΩ and R_2 = 10 kΩ
  - $V_{out} \quad \alpha \quad \frac{R_1}{R_1 + R_2} = \frac{10}{10 + 10} = 0.5$
- So, $2R_1$ = 20 kΩ and $\frac{1}{2}R_2$ = 5 kΩ
  - $V_{out} \quad \alpha \quad \frac{2R_1}{2R_1 + \frac{1}{2}R_2} = \frac{20}{20 + 5} = 0.8$
- Therefore, doubling R_1 and halving R_2 leads to an increase in V_{out}

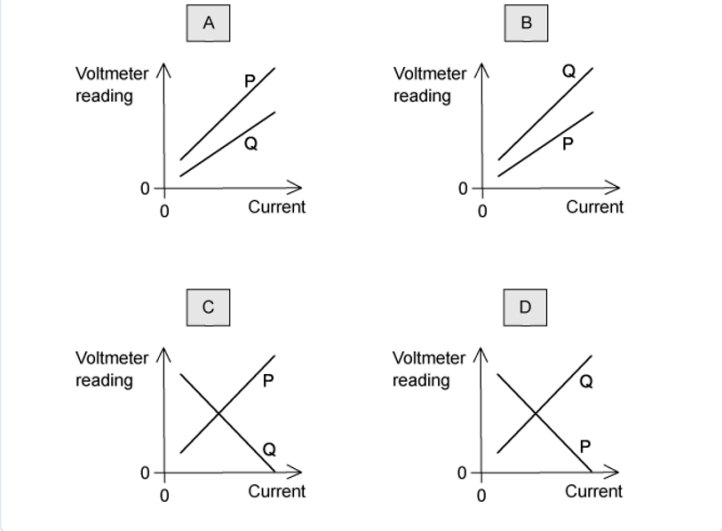
Question 17 Medium [1 mark]

A 12 V battery is in series with an ammeter, a 2 Ω fixed resistor and a 0 – 10 Ω variable resistor. High-resistance voltmeters P and Q are connected across the variable resistor and the fixed resistor, respectively, as shown.



The resistance of the variable resistor is changed from its maximum value to zero.

Which graph shows the variation with current of the voltmeter readings?



A

B

C

D

Mark Scheme & Explanation:

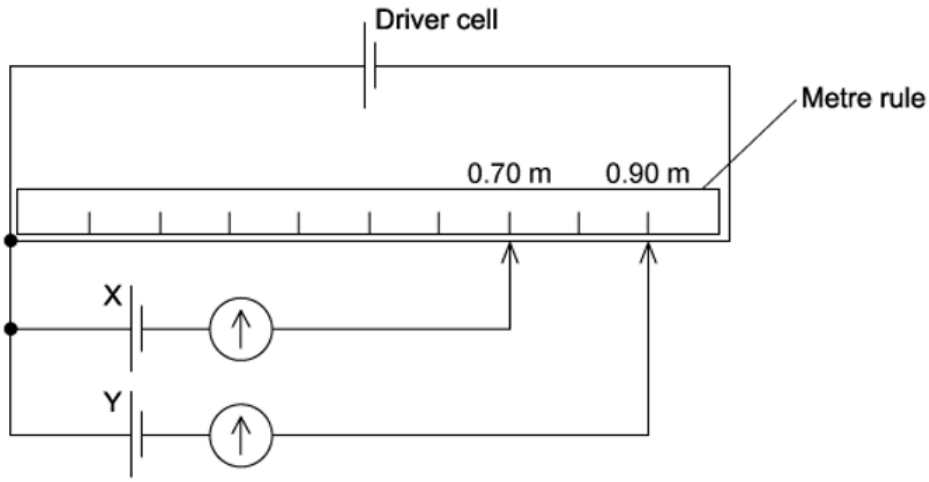
*The correct answer is **D** because:*

- Since the variable resistor and the fixed resistor are in series, the total e.m.f in the circuit is divided between them
- The potential difference across each of the resistors may be obtained from the potential divider equation : $V_P = V \times \frac{R_P}{R_P + R_Q}$
- Let's say R_P is the resistance of the variable resistor and R_Q is the resistance of the fixed resistor
- As R_P is decreased, the p.d across it decreases $V = IR$
    - When R_P = 0, V_P = 0, therefore, the voltmeter reading of P should give zero - option D shows this
    - Decreasing R_P should also cause the p .d. across the fixed resistor to increase - this is also shown by option **D**

A, B & C are incorrect because as R_P decreases, the p. d. across it also decreases, whereas these graphs all show the p .d. increasing

Question 18 Medium [1 mark]

A potentiometer is used as shown to compare the e.m.f. of two cells.



The balance points for cells X and Y are 0.70 m and 0.90 m respectively.

If the e.m.f. of cell X is 1.1 V, what is the e.m.f. of cell Y ?

A 0.69 V

B 0.86 V

C 0.99 V

D 1.4 V

Mark Scheme & Explanation:

*The correct answer is **D** because:*

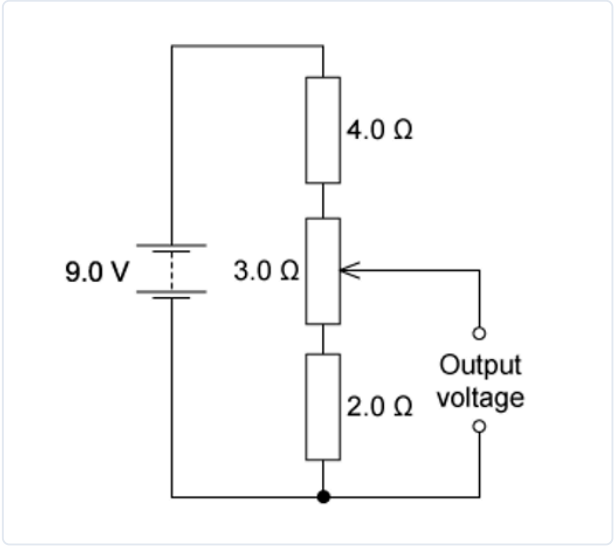
- At the balance length, the p.d. across the wire is equal to the e.m.f. of the cell
- The p.d is directly proportional to the length of the wire i.e. $V \propto L$
  - This is because p.d is proportional to work done, which is in turn proportional to distance i.e. $V \propto W \propto d$
- Cell X: $1.1\ \text{V} \propto 0.70\ \text{m}$ **(1)**
- Cell Y: $E \propto 0.90\ \text{m}$ **(2)**
- To find the ratio, divide equation **(2)** by **(1)**:

- $\frac{E}{1.1} = \frac{0.90}{0.70}$
- $E = \frac{0.90}{0.70} \times 1.1 = 1.41 \text{ V}$

- Therefore, the e.m.f. of cell Y is 1.4 V

Question 19 Medium [1 mark]

A potential divider circuit consists of fixed resistors of resistance 2.0 Ω and 4.0 Ω connected in series with a 3.0 Ω resistor fitted with a sliding contact. These are connected across a battery of e.m.f. 9.0 V and zero internal resistance, as shown.



What are the maximum and minimum output voltages of this potential divider circuit?

	Maximum voltage / V	Minimum voltage / V
A	4.0	2.0
B	5.0	2.0
C	9.0	0
D	9.0	2.0

A

B

C

D

Mark Scheme & Explanation:

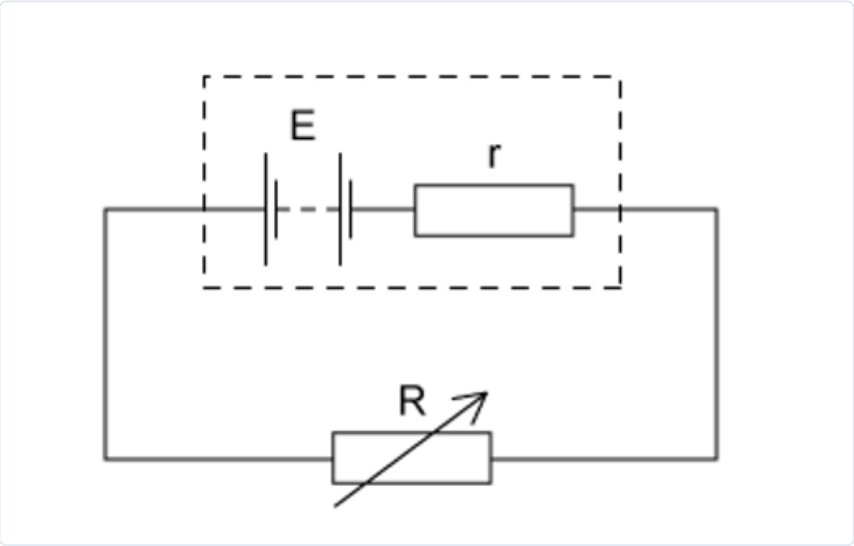
*The correct answer is **B** because:*

- The potential at the top of the circuit (+ terminal) is 9 V, and the potential at the bottom of the circuit (- terminal) is 0 V
- There will be a p. d. across each of the fixed resistors

- Potential divider equation for three resistors: $V_{out} = V \times \frac{R_1 + R_2}{R_1 + R_2 + R_3}$
- The output voltage is maximum at the maximum value of the variable resistor = 3.0 Ω
  - Maximum output voltage $= 9.0 \times \frac{2+3}{2+3+4} = 5.0 \text{ V}$
- The output voltage is minimum when the minimum value of the variable resistor is taken = 0 Ω
  - Minimum output voltage $= 9.0 \times \frac{2+0}{2+3+4} = 2.0 \text{ V}$

Question 20 Medium [1 mark]

A battery with e.m.f. *E* and internal resistance *r* is connected in series with an external variable resistor.



The value of the external resistance *R* is slowly increased from zero.

Which statement is correct ? (Ignore any temperature effects.)

A the potential difference across the external resistance decreases

B the potential difference across the internal resistance increases

C the power dissipated in *r* increases and then decreases

D the power dissipated in *R* increases and then decreases

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Ohm's law states $V = IR$, so, as the resistance of a component increases the p. d. across it increases
- So, as the value of external resistance *R* is increased, the p. d. across it increases, while the p. d. across the internal resistance decreases
- The external resistance *R* and the internal resistance r form a potential divider
- Power dissipated in external resistor $= I^2 R$
    - Initially, the value of *R* is zero, so the power dissipated across it is also zero
    - Similarly, if the value of *R* is infinite, the power dissipated is also zero as the current will then also be zero

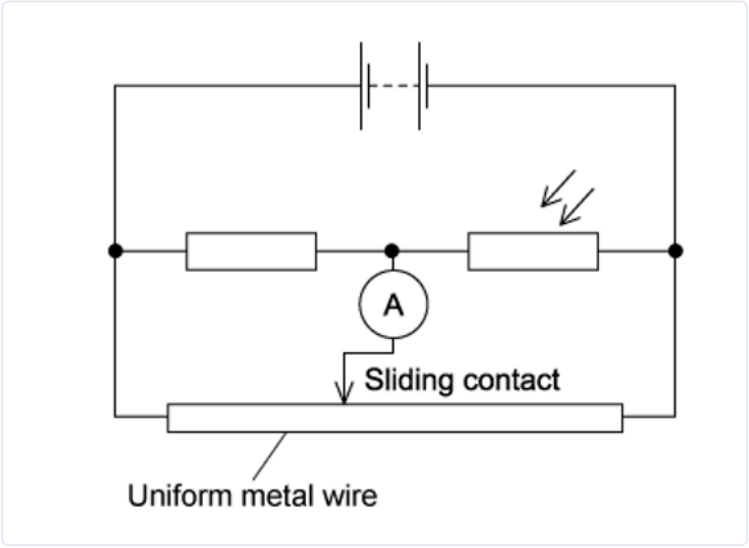
- So, the power dissipated must increase and then decrease as *R* increases from zero
    - As the value of *R* increases, the current I decreases and hence the p. d. across the external resistor increases
    - The p. d. across the internal resistor *r* decreases and the power dissipated in *r* (given by $I^2 R$) decreases
    - The maximum power dissipated across R occurs when *R* = *r*
- Overall, the only correct statement is that the power dissipated in R increases and then decreases

A & **B** are incorrect because the sum of p .d. in a circuit is equal to the e .m .f. in the circuit, so as the p .d. of one component rises, the other will fall

C is incorrect because as the value of *R* increases, the p d. across the external resistor increases, so the p. d. across the internal resistor decreases and the power dissipation across it also decreases

Question 21 Hard [1 mark]

In the potentiometer circuit shown, the reading on the ammeter is zero.



The light-dependent resistor (LDR) is then covered up, and the ammeter gives a non-zero reading.

Which change could return the ammeter reading to zero?

A decrease the supply voltage

B increase the supply voltage

C move the sliding contact to the left

D move the sliding contact to the right

Mark Scheme & Explanation:

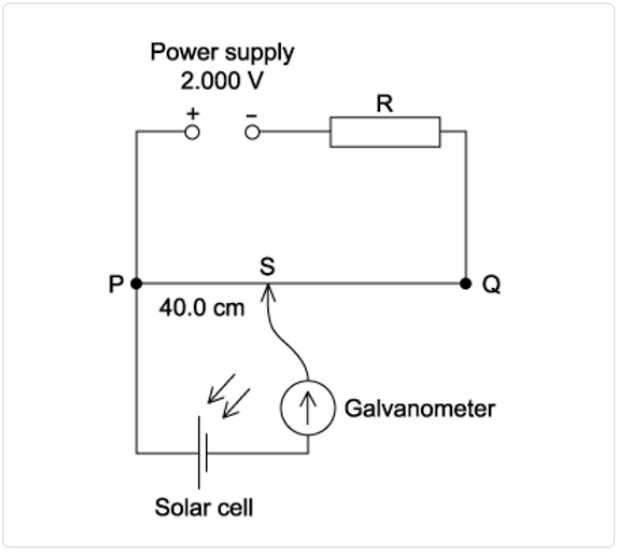
*The correct answer is **C** because:*

- For the ammeter to read zero, the potential difference across it should be zero; hence, the terminals of the ammeter are at the same potential
- The resistance of an LDR decreases when the light intensity on it increases, so, as the LDR is covered up, the light intensity decreases - this causes its resistance to increase
- Considering the resistor and the LDR in the middle section, an increase in the resistance of the LDR causes the p. d. across it to increase $V = IR$
    - So, the terminal connected between the LDR and the resistor is now at a higher potential than before

- To return the ammeter reading to zero, the sliding contact should be moved in such a way that it is at a higher potential (and equal to the new potential of the other terminal of the ammeter)
- The right end of the uniform metal wire is connected to the negative terminal of the supply and so, is at a potential of zero
- The left end of the wire is connected to the positive terminal of the supply and is at the maximum potential (equal to the e. m .f of the supply)
- So, as we move from right to left, the potential increases
    - Therefore, to increase the potential of the sliding contact, we need to move it to the left

Question 22 Hard [1 mark]

A power supply and a solar cell are compared using the potentiometer circuit shown.



The e.m.f. produced by the solar cell is measured on the potentiometer.

The potentiometer wire PQ is 100.0 cm long and has a resistance of 5.00 Ω. The power supply has an e. m .f. of 2.000 V, and the solar cell has an e .m. f. of 5.00 mV.

Which resistance *R* must be used so that the galvanometer reads zero when PS = 40.0 cm ?

A 395 Ω

B 795 Ω

C 995 Ω

D 1055 Ω

Mark Scheme & Explanation:

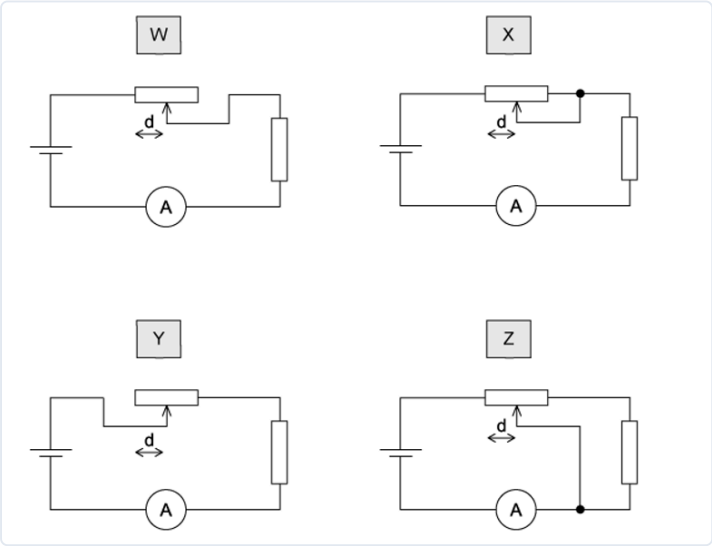
*The correct answer is **B** because:*

- A galvanometer is a type of sensitive ammeter an instrument for detecting electric current
- In order for the galvanometer to read zero, the e. m .f of the solar cell must be equal to the p .d. such that the p .d. across the galvanometer is zero
- The potential divider equation is given by
  - $V_{out} = V_0 \frac{R_1}{R_1 + R_2}$

- When PS = 40.0 cm, the p. d. across the length of wire, PS is given by
  - $2.000 \quad \text{V} \times \frac{0.400 \times 5.00}{R + 5.00}$
- So, by equating this p .d. to the e. m .f. of solar cell (5.00 mV), resistance R can be obtained by rearranging the equation:
  - $2.000 \times \frac{0.400 \times 5.00}{R + 5.00} = 5.0 \times 10^{-3} \quad \text{V}$
  - $2.000 \times 0.400 \times 5.00 = 5.0 \times 10^{-3} R + 500$
  - $4.00 = 5.0 \times 10^{-3} R + 0.025$
  - $R = \frac{4.00 - 0.025}{5.0 \times 10^{-3}} = \frac{3.975}{5.0 \times 10^{-3}} = 795 \quad \Omega$
- Therefore, the resistance R is equal to 795 Ω

Question 23 Hard [1 mark]

The diagrams show the same cell, ammeter, potentiometer and fixed resistor connected in different ways.



The distance *d* between the sliding contact and a particular end of the potentiometer is varied. The current measured is then plotted against the distance *d*.

For which two circuits will the graphs be identical ?

A W and X

B W and Y

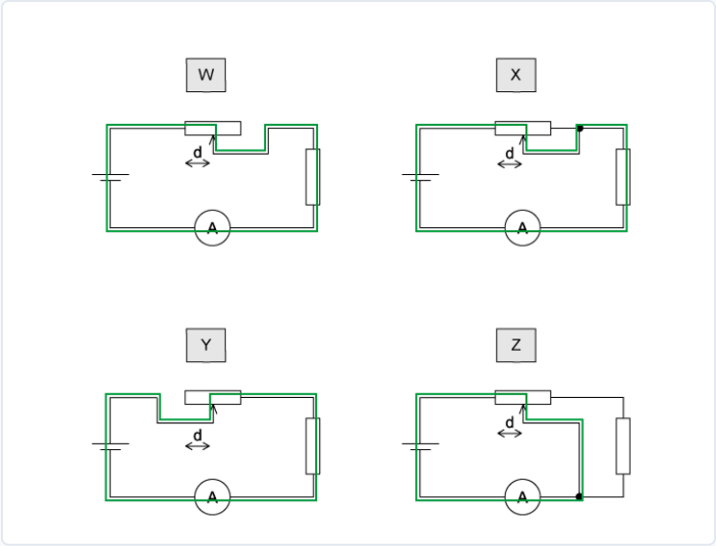
C X and Y

D Y and Z

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A potentiometer is different than a regular resistor in that the potentiometer can allow for different values of resistance
- Current always chooses the easiest way, that is, the path is the smallest resistance
- The best way to approach this question is to draw out the path of the current in each circuit
- As we can see from the diagram, the path of the current is the same in W and X; therefore, they will give identical graphs

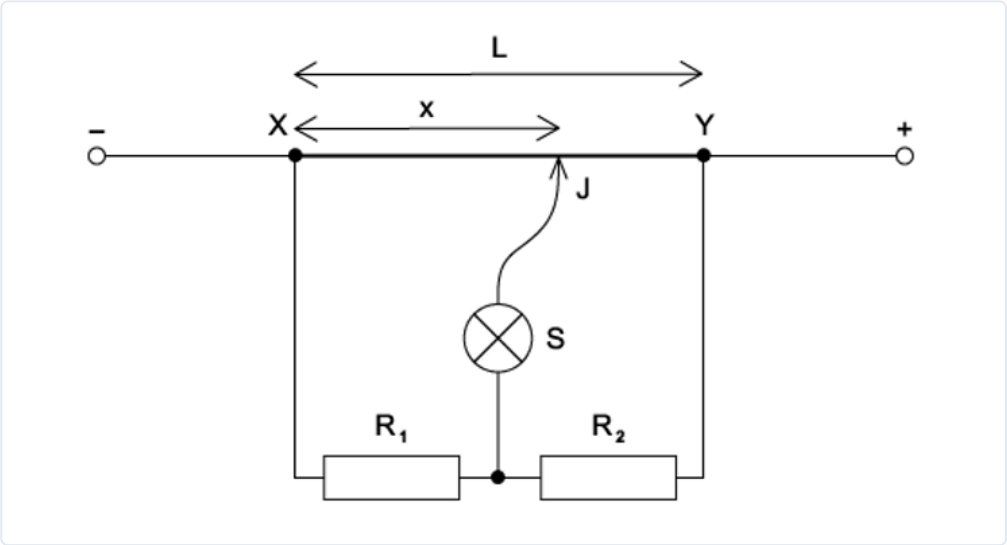


B & **C** are incorrect because in arrangement Y, the path other than the distance *d* on the wire of the potentiometer is connected to the circuit. So, this arrangement is different from those of W and X. Even if the circuit is complete, the results obtained on a graph would not be identical to that of W or X

D is incorrect because in arrangement Z, current flows from the positive terminal of the battery, then along a distance *d* on the potentiometer, but here, the current does not flow through the fixed resistor; hence it would not give the same graph as in Y

Question 24 Hard [1 mark]

In the circuit shown, XY is a length *L* of uniform resistance wire. R_1 and R_2 are unknown resistors. J is a sliding contact that joins the junction of R_1 and R_2 to points on XY through a small signal lamp S.



To determine the ratio $\frac{V_1}{V_2}$ of the potential differences across R_1 and R_2, a point is found on XY at which the lamp is off. This point is at a distance *x* from *X*.

What is the value of the ratio $\frac{V_1}{V_2}$?

A $\frac{L}{x}$

B $\frac{x}{L}$

C $\frac{L-x}{x}$

D $\frac{x}{L-x}$

Mark Scheme & Explanation:

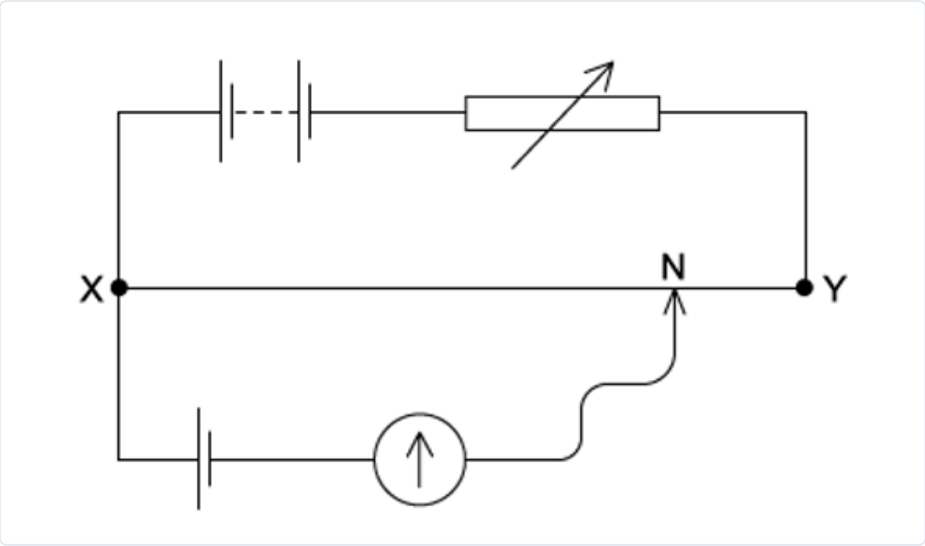
*The correct answer is **D** because:*

- For the lamp to be off, the potential difference across it should be zero so that no current flows through it
  - This means, the potential at the junction of R_1 and R_2 should be equal to the potential at the sliding contact J
- The p. d. across R_1 is V_1 and the p .d. across R_2 is V_2, so, V_1 should be equal to the p. d. across XJ and V_2 should be equal to the p .d. across JY

- Since point X is connected to the negative terminal of the supply, it can be considered to be at a potential of 0, while point Y is at the maximum potential, equal to the e .m. f. of the supply
- The wire XY, together with the sliding contact J, can be considered as a potential divider
- Resistance *R* of the wire $= \frac{\rho L}{A}$ and from Ohm's Law$V = IR$, so, the p. d. across the wire is proportional to the length of wire $V \quad \alpha \quad L$
- A length of *L* corresponds to a p. d. equal to the e .m .f. *E* of the supply and a length of *x* corresponds to a p. d. equal of $\frac{E}{L} \times x = \frac{x}{L} \times E$
- Since X is at a potential of zero, point J would be at a potential of:
  - $V_1 = \frac{x}{L} \times E$ - $0 = \frac{x}{L} \times E$
- Now, the part JY of the wire corresponds to a length (*L* – *x*) of the wire. A length of (*L* – *x*) corresponds to a p.d. equal to
  - $V_2 = \frac{E}{L} \times L$ - $x = \frac{L - x}{L} \times E$
- Therefore, the ratio $= \frac{V_1}{V_2} = \frac{x}{L} E \div \frac{L - x}{L} E = \frac{x}{L - x}$

Question 25 Hard [1 mark]

In the potentiometer circuit below, the moveable contact is placed at N on the bare wire XY, such that the galvanometer shows zero deflection.



The resistance of the variable resistor is now increased.

What is the effect of this increase on the potential difference across the wire XY and on the position of the moveable contact for zero deflection ?

	Potential difference across XY	Position of moveable contact
A	increases	nearer to X
B	increases	nearer to Y
C	decreases	nearer to X
D	decreases	nearer to Y

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Let's say the resistance of the variable resistor = *R*
    - e. m .f of driver cell = p .d. across variable resistor + p .d. across wire XY

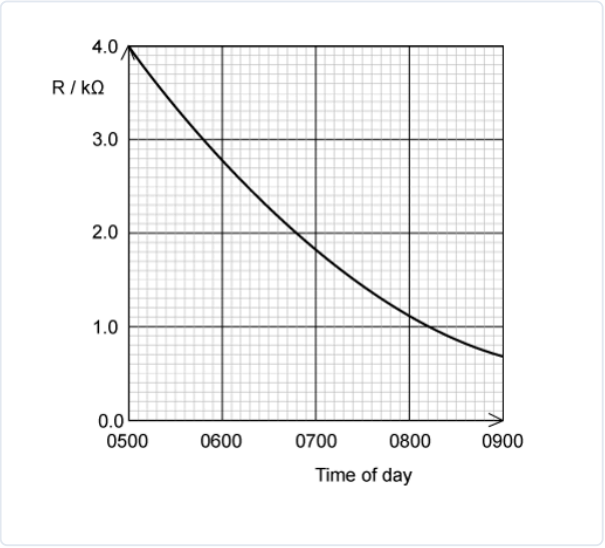
- As the resistance *R* of the variable resistor is increased, the p. d. across it also increases $V = IR$ and hence, the p. d. across the wire XY decreases
- For the galvanometer to show zero deflection, the potential difference across it should be zero, i.e. no current flowing through it
    - This means the potential at one side of the galvanometer should be equal to the potential on the other side, making the potential difference zero
- Initially, before the resistance of the variable resistor is increased, the galvanometer showed zero deflection
    - So, the potential difference across it was zero and the p. d. across XN was equal to the e .m .f. of the cell
- But an increase in the resistance of the variable resistor causes the p. d. across XY to decrease
    - So, to cause the galvanometer to show zero deflection again, the p .d. across wire XN should be increased so that it becomes equal to the e .m .f of the cell again
- Using the equations $V = IR$ and $R = \frac{\rho L}{A}$, we can see that $V \quad \alpha \quad L$
    - Therefore, to increase the p. d. the length must be increased
    - This is done by moving the contact closer to end Y

A & **B** are incorrect because when the resistance *R* increases, the p. d. across it also increases and hence, the p. d. across the wire XY decreases

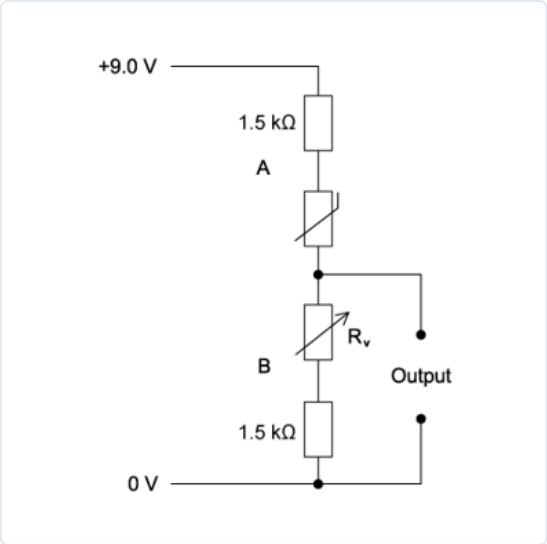
C is incorrect because to cause the galvanometer to show zero deflection again, the moveable contact must move nearer to Y since $V \quad \alpha \quad L$

Question 26 Hard [1 mark]

The graph below shows how resistance *R* of a thermistor varies during part of a day.



The thermistor is connected to the potential divider circuit as shown.



At 0730, an output of 6.0 V is required.

What value of the variable resistor should be set at this time?

A 1.5 kΩ

B 3.0 kΩ

C 4.5 kΩ

D 6.0 kΩ

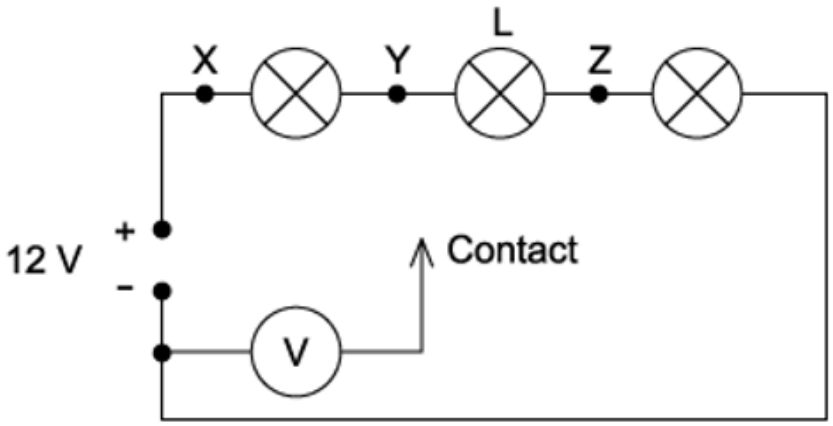
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Section A is made up of a 1.5 kΩ fixed resistor and the thermistor
- Section B is made up of the variable resistor and another 1.5 kΩ fixed resistor
  - If the output is to be set at 6.0 V, then across the section A, the voltage drop needs to be 3.0 V
- From the graph, at 0730, the resistance of the thermistor = 1.5 kΩ
- From Ohm's Law: $V = IR$
- The current flowing through section A:
  - $I = \frac{V}{R} = \frac{3.0 \quad V}{1500 + 1500} = 0.001 \quad A$
- Similarly, the same current flows through section B:
  - $6.0 \quad V = IR = 0.001 R_v + 1500$
  - $0.001 \quad R_V = 6.0 - 1500 \times 0.001 = 4.5$
  - $R_V = \frac{4.5}{0.001} = 4500 \quad \Omega = 4.5 \quad k\Omega$
- Therefore, at 0730 the variable resistor should be set to 4.5 kΩ

Question 27 Hard [1 mark]

The contact is connected one at a time to points X, Y and Z to test the circuit. The lamps do not light up because lamp L is blown.



Which of the following are the readings of the voltmeter when the contact is connected to each of the points?

	X	Y	Z
A	12 V	8 V	4 V
B	8 V	8 V	0 V
C	12 V	12 V	0 V
D	8 V	12 V	4 V

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

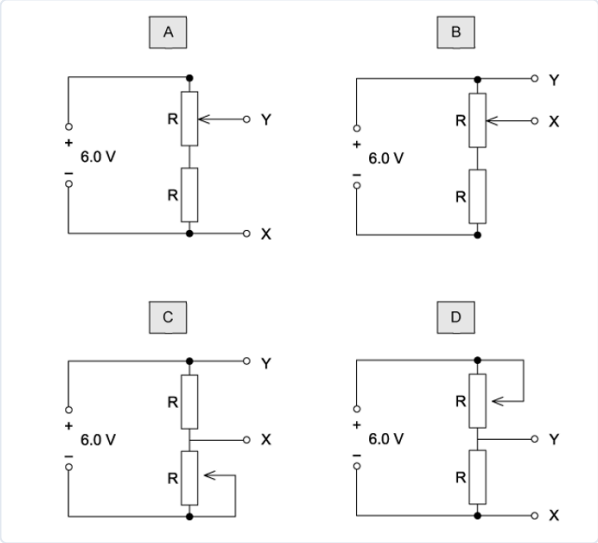
- For point X, the voltmeter is merely measuring the voltage between the terminals of the battery as there are no electrical components in between
    - Hence, the reading is that of the voltage of the battery, which is 12 V
- For point Y, there is a lamp in between the ends of the voltmeter

- However, due to the broken filament of lamp L, no current flows in the circuit
- According to $V = IR$, the voltage across the lamp between X and Y will be zero
    - Hence, the voltage measured by the voltmeter will still be that between the terminals of the battery, which is 12 V
- Finally, placing the contact at point Z has the same effect as placing it at the negative terminal, so, the voltmeter will read the voltage between the negative terminal and itself
    - Hence, the reading registered will be 0 V

Question 28 Hard [1 mark]

A potential divider has a constant supply of 6.0 V, as shown in the diagrams.

Which circuit will provide a potential difference between X and Y that can be varied between zero and 3.0 V ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Since the two resistors are in series, the total e.m.f in the circuit is divided between them
- The variable resistor controls the range of the output voltage, so X and Y must be placed across the variable resistor
- Consider option B:
  - When the resistance of the top resistor is maximized to R, the potential difference between X and Y = 3.0 V
  - When the resistance of the top resistor is minimized to 0, the potential difference between X and Y = 0 V
- Therefore, option **B** is correct

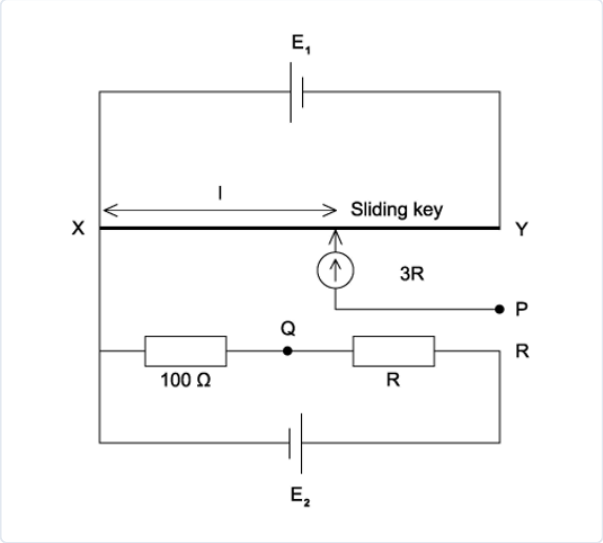
A is incorrect because the range of output voltage of this circuit would be 3 - 6 V as it includes both resistors

C & **D** are incorrect because the variable resistor controls the range of the output voltage, so X and Y must be placed across the variable resistor

Question 29 Hard [1 mark]

The resistance *R* of an unknown resistor is compared with a fixed 100 Ω resistor using the circuit as shown in the diagram.

The balance length *l* is 30.0 cm and 48.3 cm when P is connected to Q and R, respectively.



The length of the wire XY is 100 cm and has a resistance of 40 Ω. The two cells E_1 and E_2, have negligible internal resistances.

What is the value of resistance *R* ?

A 38 Ω

B 47 Ω

C 61 Ω

D 161 Ω

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- When P is connected to Q, the potential difference across the balance length is equal to the potential difference across the 100 Ω resistor
  - $\frac{l}{XY} \times E_1 = \frac{100}{100+R} \times E_2$ **(1)**
- When P is connected to R, the potential difference across the balance length is equal to the potential difference across the 100 Ω resistor and the unknown resistor (which is the same as the e. m .f. as cell E)

- $\frac{l'}{XY} \times E_1 = E_2$ **(2)**

- Substitute equation **(2)** into **(1)**
  - $\frac{l}{XY} \times E_1 = \frac{100}{100+R} \times \frac{l'}{XY} \times E_1$
  - $\frac{30.0}{XY} \times E_1 = \frac{100}{100+R} \times \frac{48.3}{XY} \times E_1$
- The E_1 and XY terms cancel out, so we have:
  - $30.0 = \frac{100}{100+R} \times 48.3$
  - $100 + R = \frac{100 \times 48.3}{30.0} = 161$
  - $R = 161 - 100 = 61 \quad \Omega$
- Therefore, the resistance *R* of the unknown resistor is 61 Ω